

Difference between JPA, Hibernate and Spring Data JPA

JPA, Hibernate and Spring Data JPA are not different technologies that compete with each other. They actually work together, and each one has a different job.

JPA (Java Persistence API) is a specification that tells developers how Java objects should be stored and retrieved from a database. It only defines the rules and APIs, but it does not contain any code to perform these operations. This means JPA alone cannot communicate with the database.

Hibernate is an ORM framework that implements the JPA specification. It does the actual work of converting Java objects into SQL queries and storing or retrieving data from the database. While using Hibernate directly, the developer has to create sessions, start and commit transactions, handle exceptions and close the session manually, which increases the amount of code.

Spring Data JPA is built on top of JPA and works with implementations like Hibernate. It does not replace Hibernate. Instead, it makes development much easier by reducing the amount of code we write. It provides interfaces like `JpaRepository`, which already contain common database operations such as save, delete and find. Spring also manages transactions using the `@Transactional` annotation, so the developer can focus more on the business logic instead of writing repetitive database code.

In simple terms, JPA defines the rules, Hibernate follows those rules and performs the database operations, while Spring Data JPA makes using Hibernate much simpler by reducing boilerplate code and providing ready-made repository methods.